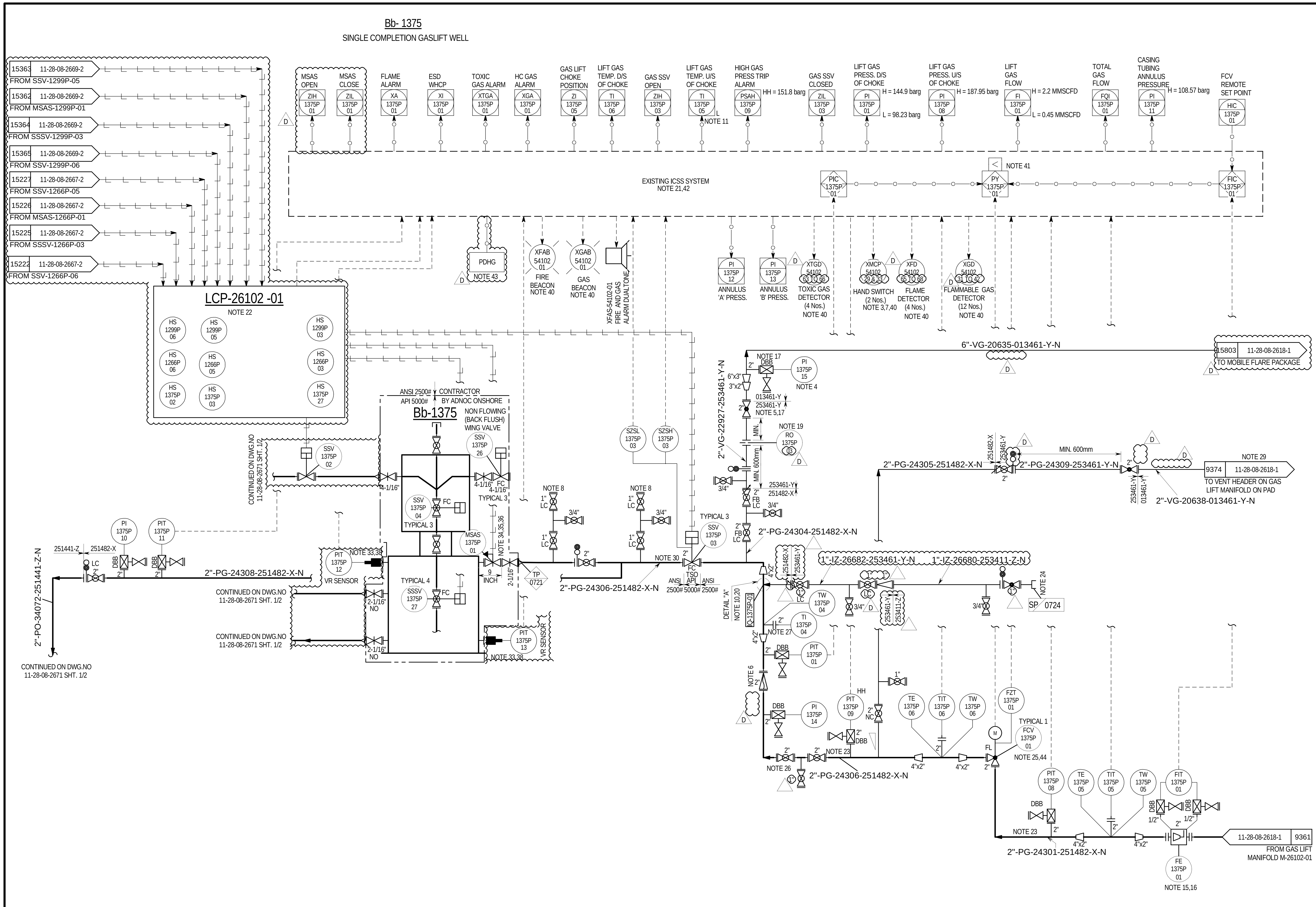

NOTES

1. ALL EQUIPMENT, INSTRUMENTS & PIPING SPECIALITY TAG NUMBERS ON THIS DRAWING SHALL BE PREFIXED WITH AREA CODE (11) BAB AND PLANT AREA CODE (28) FOR SINGLE WELLS.
2. CASING TO CASING AND/OR CASING TO TUBING BLEED OFF OPERATION IS MANUAL UNDER OPERATING PROCEDURES. WHEN ANNULUS HIGH PRESSURE IS DETECTED, OPERATOR TO ENSURE THAT THERE IS NO FLUID STAGNANCY.
3. SWITCHES ON WELLHEAD PANEL TO OPEN/CLOSE GAS LIFT ESDV AT B/L, SDV, WING VALVES AND SSV. SWITCH ALSO PROVIDED FOR OPENING OF SUBSURFACE SAFETY VALVE SSSV AND KILL WING VALVE.
4. PRESSURE GAUGE TO BE VISIBLE FROM GLOBE VALVE.
5. GLOBE VALVE TO BE THROTTLED TO LIMIT THE VENTING RATE SUCH THAT DOWNSTREAM PRESSURE DOES NOT EXCEED 3.0 barg.
6. NON SLAM NOZZLE TYPE CHECK VALVE.
7. ESD PUSH BUTTONS PROVIDED. TWO ON WHCP, REST TO BE PROVIDED ON ACCESS GATES.
8. TAPPING POINT TO CONNECT THE ECHO METER OR ACOUSTIC GUN FOR ACOUSTIC SURVEY RESULTS.
9. DELETED. D
10. INJECTION QUILL TO BE PROVIDED REFER DETAIL "A"OF LEGEND SHEET 11-99-08-2601.
11. LOW ALARM TO BE SET @ -25°C. TO BE READ DURING PRESSURIZATION OF WELLHEAD GAS LIFT PIPING.
12. DELETED.
13. DELETED. D
14. DELETED.
15. STRAIGHT PIPE LENGTH OF 3D UPSTREAM AND 1D DOWNSTREAM OF V-CONE FLOWMETER.
16. GAS FLOW MEASUREMENTS TO HAVE COMPENSATION AND TO BE CONFIGURED FOR BOTH LEAN & RICH GAS COMPOSITIONS.
17. VALVE TO BE PROVIDED WITH EXTENDED BONNET.
18. DELETED. D
19. MULTISTAGE RO. ADEQUATE SUPPORT SHALL BE PROVIDED DOWNSTREAM OF RO PIPING TO AVOID VIBRATION DUE TO HIGH VELOCITY.
20. STANDARD INJECTION NOZZLE AND ACCESS FITTING (COSASCO OR EQUIVALENT FITTING) REFER DETAIL "A" IN LEGEND SHEET NO. P11671-11-99-08-2601 FOR TYPE & DETAILS.
21. FOR DETAILS OF PLC SIGNALS REFER I/O LIST.
22. FOR EXISTING WELLS, 1 NO. ADDON GAS LIFT PANEL FOR GL & Y-TYPE XMAS TREE CONSUMERS (2 GL + 2 SPARE BLANK PLATE SLOT) SHALL BE CONSIDERED & INTEGRATED WITH EXISTING ICSS.
23. LINE TO BE ADEQUATELY SUPPORTED FOR TWO PHASE FLOW DURING START-UP.
24. PROVISION FOR CHEMICAL INJECTION POINT. PERMANENT CI INJECTION IS NOT CONSIDERED. IN CASE REQUIRED, A SMALL SEPARATE CI PACKAGE OR MULTIHEAD PUMP NEED TO BE INCLUDED AND SHALL BE SOLAR POWERED.
25. CHOKE VALVE IS MOTORIZED WITH REMOTE OPERATION.
26. MONO BLOCK DOUBLE BLOCK AND BLEED ARRANGEMENT.
27. TEMPERATURE INDICATOR HAS TO BE VISIBLE DURING MANUAL DEPRESSURIZATION OPERATION.
28. DELETED. D
29. THE VENT LINE FROM THE WELL TO BE TIED IN TO A COMMON VENT HEADER AT GAS LIFT MANIFOLD ON WELLBAY I.E. VENT TO ATMOSPHERE AT SAFE LOCATION. MINIMUM LENGTH DOWNWIND FROM OPERATING AREA AND VENT HEIGHT SHALL BE CONFIRMED BASED ON DISPERSION ANALYSIS.
30. MATING FLANGE FOR SSV SHALL BE API5000 RATED MATCHING WITH PIPING SCHEDULE.
31. DELETED.
32. DELETED.
33. MINIMUM 5FT CLEARANCE TO BE PROVIDED IN PIPING ARRANGEMENT FOR VR SENSOR INSTALLATION.
34. ADEQUATE SPACE FOR MSAS INSTALLATION D/S OF GAS LIFT SSV TO BE PROVIDED.
35. MASTER SURFACE ANNULUS SAFETY VALVE IS SUPPLIED BY ADNOC ONSHORE AND SHALL BE INTERFACED BY CONTRACTOR WITH THE WHCP AND WELLHEAD PLC.
36. MSAS REQUIRE HYDRAULIC OIL TO REMAIN IN OPEN POSITION AND AVOID CHATTERING. MSAS SHALL HAVE ITS INDEPENDENT HYDRAULIC LINE AND NOT TO BE SHARED WITH SSSV HYDRAULIC SUPPLY FOR THE NEW WELLS. INCASE OF EXISTING MODIFIED WELL TO ACCOMMODATE MSAS, A COMMON HYDRAULIC SUPPLY LINE CAN BE SHARED AMONG MSAS AND SSSV SUPPLY AND WILL SERVE AS A BUFFER ACCUMULATOR FOR MSAS ACTUATION ALLOWING TO OPEN MSAS FOR BLEEDING WHILE CLOSING THE SSSV. REFER TO DETAIL-B FOR THE CASE OF EXISTING WELLS REVAMP WITH MSAS MODIFICATION.
37. DELETED.
38. VR SENSORS & MSAS VALVE ARE PROVIDED AS PART OF WELLHEAD SCOPE OF SUPPLY (BY ADNOC ONSHORE). CONTRACTOR IS RESPONSIBLE FOR INTERFACING THE SIGNALS FROM THESE SENSORS WITH WELLHEAD PLC.
39. DELETED. D
40. THE NUMBER OF ESD PUSH BUTTON AND NUMBER OF TOXIC GAS DETECTORS SHALL BE DECIDED BASED ON FIRE & GAS PHILOSOPHY AND FIRE & GAS DETECTORS LAYOUTS. MANUAL ESD PUSH BUTTON IS PROVIDED IN WHCP.
41. WHEN THE FLOW RATE IS LESS THAN THE SET POINT, THE CHOKE VALVE OPENS TO MAINTAIN CONSTANT LIFT GAS FLOW RATE AND VICE VERSA WITH AN INPUT FROM THE DOWNSTREAM PRESSURE TRANSMITTER.
42. ALL SIGNALS OF THIS WELL WILL BE CONFIGURED IN THE EXISTING ICSS SYSTEM FOR THE CURRENT PROJECT SCOPE.
43. DOWNHOLE PRESSURE GAUGE IS FREE ISSUED BY COMPANY, AND SHALL BE INSTALLED AND INTERFACED WITH PLC AS PART OF PROJECT SCOPE.
44. CHOKE VALVE IS ADJUSTABLE WITH POSITIVE LOCKING DEVICE.

D

LEGENDS

- BY OTHERS
- OUT OF SCOPE
- REVISION CLOUD

REV.	DATE	DR'N.	CH'D.	AP'D.	DESCRIPTION
D	18.07.2025	PT	TL/ST	AJ	ISSUED FOR DESIGN (IFD)
C	09.05.2025	PT	TL/ST	AJ	ISSUED FOR HAZOP (IFH)
B	06.03.2025	PT	TL/ST	AJ	ISSUED FOR DESIGN REVIEW (IDR)
A	31.01.2025	PT	TL/ST	AJ	ISSUED FOR REVIEW (IFR)

SCALE: N.T.S | LOCATION: BAB / HABSHAN | PROJECT No. P11671

HOLDS

1. DELETED.
2. DELETED.

THIS DRAWING IS NEWLY PREPARED FOR THIS PROJECT No. P11671

REFERENCE DRAWINGS

TITLE	DRG NO.
P&ID LEGEND SHEET	P11671-11-99-08-2601
P&ID SINGLE COMPLETION 2500# GAS LIFTED WELL HEAD BB-XX WITH HIPPS FOR HIGH H2S	P11671-11-71-08-0646

CONSULTANT / ~~CONTRACTOR / VENDOR~~ | EPC CONTRACTOR

DRAWING No. AET01342-00-3143072

PROJECT: EPC CALL OUT OF BAB ARTIFICIAL LIFT PHASE-II

DRG. TITLE: PIPING AND INSTRUMENT DIAGRAM
SINGLE COMPLETION ARTIFICIAL LIFT OIL WELL Bb-1375P
GASLIFT RATING 2500# HIGH H2S PAD 102 : BAB

ADNOC DRG.No.	AREA	P/AREA	DOC. CODE	SERIAL No.	REV.	SHT OF SHT
	11	28	08	2671	D	2/2